Name: Beth Armstrong
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Introduction

SQL functions are saved code that can be used repeatedly. Functions can be simple, returning a single value, or complex and able to create table-level changes. This paper will review the different types of functions and their uses.

When to use a SQL User Defined Function (UDF)

Every SQL Server database has a Programmability folder giving you access to system functions, as in Figure 1.

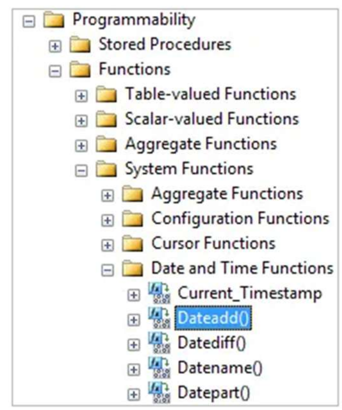

Figure 1: Functions in Programmability Folder

There are times when you need a very specific type of repeatable query, and the system doesn't provide what is needed.

A custom function, referred to as a User Defined Function (UDF) can be created when a system function does not yet exist to complete the task at hand. SQL allows users to create custom functions that are simple or complex.

Scalar, Inline, and Multi-Statement Functions

Scalar Functions
In SQL Server, a scalar function is one which returns a single value, be that a string of text, a number, or a date. Scalar functions are commonly used in Select, Where, Order By, and other clauses. Examples of some system scalar functions include UPPER, which will convert text to all uppercase or ROUND, which rounds a number up. Another example would be to select rows based on a filter parameter in the Where clause, such as last name starts with 'K.' The key characteristic is that for each row processed, the function returns exactly one value. Although scalar user-defined functions can be immensely useful, they can have a dramatic negative impact on query performance.

Inline Functions
An inline table-valued function (ITVF) is a user-defined function that returns a table as its result. Unlike scalar functions, which return a single value, inline functions can return multiple rows and columns through a single SELECT statement. Because the function

consists of only one query, SQL Server can incorporate the function directly into the execution plan, often resulting in better performance and more efficient query processing.

Inline functions are frequently used to simplify complex queries and promote code reuse. By placing commonly used query logic inside a function, developers can avoid repeating the same SQL code throughout an application. These functions can also accept parameters, allowing users to retrieve customized results based on specific criteria. For example, an inline function might return all sales transactions for a particular customer, all products within a selected category, or all employees assigned to a specific department.

One of the primary benefits of inline table-valued functions is maintainability. When business requirements change, modifications can be made within the function rather than updating multiple queries throughout the database. In addition, because SQL Server treats the function similarly to a parameterized view, inline functions generally provide better performance than more complex user-defined functions.

Multi-Statement Table-Valued Functions

While inline table-valued functions are limited to a single SELECT statement, some database operations require several processing steps before a final result can be produced. In these cases, a multi-statement table-valued function (MSTVF) provides a solution.

A multi-statement table-valued function returns a table after executing multiple SQL statements. The function begins by declaring a table variable that will hold the results. Data can then be inserted, updated, or modified within the table variable using a series of SQL statements, conditional logic, and calculations. Once all processing is complete, the populated table is returned to the user.

MSTVFs are useful when business logic is too complex to be expressed within a single query. For example, a function may need to calculate employee bonuses, aggregate sales data from multiple tables, apply conditional rules, or generate customized reports. The ability to use variables and multiple statements makes these functions highly flexible and suitable for advanced data-processing tasks.

Despite their flexibility, multi-statement table-valued functions can have performance drawbacks. MSTVFs may not perform as efficiently as inline table-valued functions, especially when working with large data sets. Developers should therefore balance the need for additional functionality against potential performance considerations.

Summary

User-defined functions are an important feature of SQL Server because they allow developers to create reusable solutions for common database operations. When built-in functions do not satisfy a particular requirement, custom functions can be developed to meet specific needs. Scalar functions are designed to return a single value, making them useful for calculations and data formatting. Inline table-valued functions return a table through a single query and are often preferred for their efficiency and performance. Multi-statement table-valued functions provide greater flexibility by supporting multiple processing steps and more complex logic before returning a result set. By understanding the capabilities and limitations of each function type, database professionals can select the most appropriate approach for improving maintainability, consistency, and overall database performance.

Citations:

Source: W3Schools, 2026, Table-valued functions in SQL accessed 5/29/2026 at https://www.wiseowl.co.uk/sql/guides/table-value-functions/table-valued-functions/

Source: W3Schools, 2026, Scalar User-Defined Functions in SQL Server accessed 5/29/2026 at https://www.wiseowl.co.uk/sql/guides/scalar-functions/scalar-udfs/